

SMART INDIA HACKATHON 2026 — PS 26100

GeM AI Compliance Platform

Software Requirements Specification (SRS)

Document Type	Software Requirements Specification
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Organization	Chennai Petroleum Corporation Limited (CPCL)
Ministry	Ministry of Petroleum & Natural Gas
Status	✓ Implemented & Live

AI-Powered · Zero-LLM · Graph-Based · Auditable

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) documents all functional and non-functional requirements of the **GeM AI Compliance Platform** — an AI-powered, integrated bid compliance verification system developed for Smart India Hackathon 2026, Problem Statement PS 26100, submitted by Chennai Petroleum Corporation Limited (CPCL) under the Ministry of Petroleum & Natural Gas.

The document serves as the authoritative reference for the implemented system, covering all features live at <http://localhost:5173> backed by a FastAPI server at <http://localhost:8000>.

1.2 Scope

The platform automates GeM procurement bid compliance verification by:

- Extracting eligibility rules from tender PDFs using AI/OCR
- Parsing and verifying GST, Udyam, and bidder documents via OCR
- Running a directed evidence graph engine to detect contradictions and fraud
- Providing a real-time compliance dashboard with risk scores
- Detecting cross-bidder cartelization via Jaccard similarity
- Supporting regional language tender documents via Bhashini (MeitY AI)
- Maintaining a SHA-256 tamper-evident audit trail

1.3 Problem Context (CPCL / PS 26100)

CPCL processes large-capital procurement tenders involving 50–200 bidders per tender, each submitting 10–15 statutory documents. Procurement officers manually cross-check information across 10+ government portals (GSTN, MCA21, Udyam, DigiLocker, etc.), resulting in:

- **250+ officer-days per tender** wasted on manual verification
- Inconsistencies and human errors in cross-portal checks
- Inability to detect document fraud (e.g., inflated turnover claims)
- No audit trail defensible in legal proceedings
- No machine-readable processing of regional language tenders

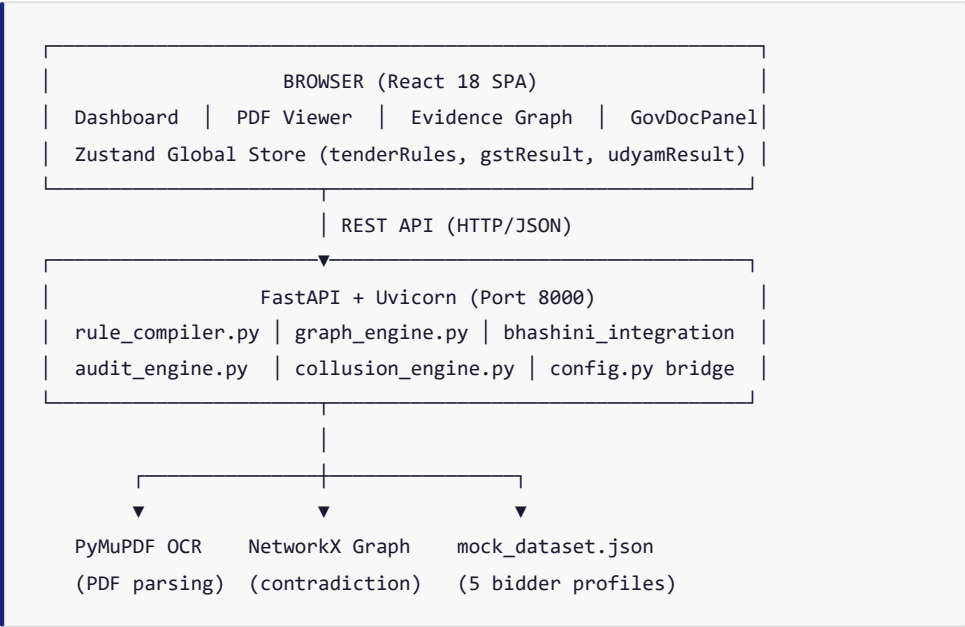
1.4 Definitions & Abbreviations

Term	Definition
GeM	Government e-Marketplace — India's national procurement portal
CPCL	Chennai Petroleum Corporation Limited
OCR	Optical Character Recognition — text extraction from PDFs
SBD	Standard Bid Document
MSME	Micro, Small and Medium Enterprise
MII	Make in India — local content mandate
EMD	Earnest Money Deposit
GFR	General Financial Rules 2017
GSTIN	GST Identification Number (15-character alphanumeric)
Bhashini	MeitY's national language translation AI platform
NIC	National Industrial Classification code
SHA-256	Secure Hash Algorithm — 256-bit cryptographic hash

2. Overall System Description

2.1 System Architecture

The system follows a **two-tier client-server architecture** with a React SPA frontend and a FastAPI asynchronous backend, connected via REST APIs over localhost.



2.2 Technology Stack

Layer	Technology	Version	Purpose
Frontend Framework	React	18	Component-based SPA UI
Build Tool	Vite	5.x	HMR + fast bundling
Styling	Tailwind CSS	3.x	Utility-first responsive design
State Management	Zustand	4.x	Lightweight global state store
Graph Visualization	ReactFlow	11.x	Interactive evidence provenance graph
Scroll Physics	Lenis	1.x	Inertia smooth scroll on dashboard
Backend Framework	FastAPI	0.111	Async REST API, auto Swagger docs

ASGI Server	Uvicorn	0.30	Production-grade async server
PDF OCR	PyMuPDF (fitz)	1.24	Text extraction from all PDFs
Graph Analytics	NetworkX	3.x	Directed evidence-claim-contradiction graph
Regional Language AI	Bhashini (MeitY)	ULCA API	OCR + translation for regional tenders
Audit Hashing	hashlib SHA-256	stdlib	Tamper-evident audit entries
Demo PDF Generation	ReportLab + Playwright	4.x / 1.x	GST, Udyam, Telugu tender demo PDFs

2.3 User Roles

Role	Actions Permitted
Procurement Officer	View dashboard, upload documents, trigger verification, review contradictions, generate show-cause notices, dismiss findings
System (Automated)	OCR extraction, graph construction, contradiction detection, score computation, audit logging
Bidder (future)	Upload documents via portal (not yet implemented)

3. Functional Requirements

FR-01: Tender Rule Extraction

Description

The system shall accept an English-language tender PDF, extract all eligibility thresholds using OCR + regex, and make them available to the verification engine and dashboard in real-time.

Inputs / Outputs

Input	Process	Output
Tender PDF (English)	PyMuPDF text extraction → DOTALL regex named capture groups	JSON array of extracted rules + updated config.active_tender_limits

Extracted Fields

- MSME turnover limit (₹ Cr)
- Sub-contracting cap (%)
- Make in India local content minimum (%)
- EMD exemption status for MSEs
- GST compliance period (months)
- Debarment prohibition clause

Acceptance Criteria

- Each extracted rule has: clause name, description, mapped_regulatory_id (UUID)
- config.active_tender_limits updated within the same request
- Dashboard auto-refreshes (tenderRules in useEffect dependency array)
- Response time < 500ms for PDFs up to 10MB

FR-02: Bidder Document Verification

Description

The system shall accept a bidder's technical bid PDF, extract all compliance claims using OCR, run those claims through the Evidence Graph Engine, and return a structured verification result.

Extracted Fields

- turnover_cr (annual audited turnover in crores)

- local_content_pct (Make in India %)
- subcontracting_pct (sub-contracting %)
- pan (PAN number — 10-char alphanumeric)
- gst_months_missed (GST filing gap count)

Acceptance Criteria

- PDF displayed in iframe immediately upon upload (before API response)
- Blue banner "Showing results from uploaded document" shown on dashboard
- "X Clear & return to dropdown" button clears upload state
- Graph engine runs contradiction detection against active_tender_limits

FR-03: GST Certificate Parsing

Description

The system shall parse a GST Registration Certificate (Form GST REG-06) PDF and extract all registration fields, flagging ineligibility conditions.

Extracted Fields & Flags

Field	Type	Flag Logic
GSTIN	15-char string	gstn_valid_format: regex [0-9]{2}[A-Z]{5}[0-9]{4}[A-Z]{1}[1-9A-Z]{1}Z[0-9A-Z]{1}
Legal Name	string	—
Constitution	enum	—
Registration Date	date	—
Registration Type	enum	composition_fail = true if type == "Composition"
is_regular_taxpayer	boolean	false if Composition/Casual/Non-Resident

UI Rendering (GovDocPanel)

- PASS badge (green) — GSTIN valid, regular taxpayer
- FAIL badge (red) — Composition dealer detected
- Red alert banner: "⚠ Composition Dealer — Ineligible for B2G Government Procurement"

FR-04: Udyam Certificate Parsing

Description

The system shall parse a Udyam Registration Certificate PDF and extract enterprise classification, flagging MSME class mismatches.

Extracted Fields & Flags

Field	Type	Flag Logic
Udyam Number	UDYAM-XX-XX-XXXXXXX	udyam_number_found boolean
Enterprise Name	string	—
Enterprise Classification	MICRO / SMALL / MEDIUM	is_micro = false if SMALL or MEDIUM
NIC Code + Description	string	—
State	string	—
Registration Date	date	—

UI Rendering (GovDocPanel)

- PASS (green) — MICRO classification confirmed
- FAIL (red) — SMALL or MEDIUM classification while claiming MICRO EMD exemption
- Red alert banner: "⚠ Enterprise is SMALL — Cannot claim MICRO EMD exemption"

FR-05: Evidence Graph Engine

Description

The system shall maintain a directed graph of all bidder claims, supporting evidence, and tender rule anchors. Contradictions between claims and evidence shall be represented as animated red EVIDENCE_CONFLICT edges.

Node Types

Node Type	Color	Example
Bidder	Black	"ACME Corp (bidder-acme-001)"
Claim	Pink	"Claimed Turnover: ₹8.5 Cr"
Evidence	Green	"MCA21: Actual Turnover ₹14.5 Cr"
Anchor:TenderRule	Purple	"MSME Turnover ≤ ₹10 Cr"

Edge Types

Edge	Meaning	Style
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MAKES_CLAIM	Bidder → Claim	Solid grey
VERIFIED_AGAINST	Claim → Evidence	Solid blue
EVIDENCE_CONFLICT	Evidence → Claim	Animated red dashed
ANCHORED_TO	Claim → TenderRule	Solid purple

Contradiction Scenarios Detected

1. **MCA21 Turnover Mismatch** — Bidder claimed < actual MCA21 turnover
2. **MII Local Content Shortfall** — local_content_pct < active_tender_limits["mii"]
3. **Sub-contracting Breach** — subcontracting_pct > active_tender_limits["subcontract"]
4. **GST Filing Gap** — gst_months_missed > 0
5. **Time-Travel Fraud** — GST filing date before company incorporation date
6. **Debarment Active** — debarred = true in bidder record

FR-06: Compliance Dashboard

Description

The system shall display a real-time compliance dashboard for any selected bidder, showing all verification results in a single unified view.

Dashboard Components

Component	Description
Bidder Selector Dropdown	5 demo bidder profiles — selection triggers auto-verification
Hard Filter Row	PAN Active / GST Active / Not Debarred — instant PASS/FAIL
Compliance Score	0–100 numeric score with colour-coded risk level
Risk Badge	LOW / MEDIUM / HIGH / CRITICAL based on score thresholds
Evidence Graph	ReactFlow interactive graph — column layout, zoom, pan
GovDocPanel	GST + Udyam OCR results — field-level PASS/WARN/FAIL badges
Contradiction Panel	All CONFLICT edges listed as actionable cards
Audit Log	SHA-256 hash per action — timestamped, tamper-evident

FR-07: Contradiction Detection & Review

Description

Each detected contradiction shall be surfaced as an actionable card with two officer actions.

Action	Effect
Dismiss Finding	Removes contradiction card; logs "DISMISSED" audit entry with SHA-256
Show-Cause Notice	Generates formatted notice text; logs "SHOW_CAUSE_ISSUED" audit entry

FR-08: Anti-Cartel / Collusion Detection

Description

The system shall scan all bidders in a tender for cartelization signals using cross-bidder analysis.

Detection Methods

- **Jaccard Similarity** — similarity of claimed values across bidders (turnover, local content, subcontracting)
- **Shared NIC Code** — multiple bidders with identical NIC codes flagged
- **Sub-contracting Ring** — bidder A subs to B who subs to A pattern

Legal guardrail: All collusion signals are presented with disclaimer — "Investigative Lead Only. Final qualification decision rests with Procurement Officer."

FR-09: Regional Language Tender — Bhashini Pipeline

Description

The system shall accept tender documents in Indian regional languages, translate them to English via Bhashini (MeitY AI), and extract compliance rules from the translated text.

Supported Languages

Telugu (te), Malayalam (ml), Hindi (hi), Tamil (ta), Kannada (kn), Bengali (bn)

Processing Pipeline

1. Upload regional PDF to Tender Document tab in PDF viewer
2. Click "IN Bhashini Translate" button (only visible on Tender tab)
3. Select source language from dropdown
4. POST /api/v1/tenders/translate-regional

- 5. PyMuPDF tries text extraction — if text layer exists → use directly
- 6. If scanned image PDF (<50 chars) → Bhashini OCR + Translation API
- 7. Rule compiler extracts compliance thresholds from resulting text
- 8. Dashboard shows: translation source badge, preview, rules extracted count

Translation Source Badges

- ✓ BHASHINI API

 — live Bhashini ULCA API used (BHASHINI_TOKEN set)
- ⚠ SIMULATION MODE

 — no token; sandbox simulation
- PDF_TEXT_EXTRACTED

 — text-layer PDF; OCR not needed

FR-10: SHA-256 Audit Trail

Description

Every officer action shall produce a tamper-evident audit entry containing a SHA-256 hash of the action data.

Audit Entry Schema

```
{
  "timestamp": "2026-09-06T14:23:11Z",
  "action": "SHOW_CAUSE_ISSUED | DISMISSED | VERIFIED | UPLOADED",
  "bidder_id": "bidder-delta-004",
  "officer_id": "officer-001",
  "payload": { ... action-specific data ... },
  "sha256_hash": "a3f9b2c1d4e8f7a6b5c4d3e2f1a0b9c8..."
}
```

Acceptance Criteria

- Hash computed over: timestamp + action + bidder_id + officer_id + payload JSON
- Audit log displayed on dashboard — most recent entry at top
- Entries are append-only (no delete endpoint)

FR-11: PDF Viewer & Tab Navigation

Description

The right panel shall contain a resizable PDF viewer with three document tabs, each with context-aware upload buttons.

Tab	Default PDF	Upload Button	Color	Backend Endpoint
Tender Document	tender_demo.pdf	Upload Bidder Document	Grey	/verify-document

		+ IN Bhashini		
GST Certificate	gst_tender_demo.pdf	Upload GST Certificate	Orange	/parse- gst
Udyam Certificate	udyam_tender_demo.pdf	Upload Udyam Certificate	Green	/parse- udyam

4. REST API Specification

Method	Endpoint	Description	Key Response Fields
POST	/api/v1/tenders/{id}/compile-rules	OCR tender PDF → extract rules → update config	extracted_rules[], tender_id
POST	/api/v1/verify/bidder/{bid}/tender/{tid}	Run graph engine on mock dataset bidder	compliance_score, risk_level, contradictions[]
GET	/api/v1/dashboard/bidder/{id}	Return serialized graph + score + contradictions	nodes[], edges[], score, risk_level
POST	/api/v1/bidders/verify-document	OCR bidder tech bid PDF → graph engine	extracted_claims, compliance_score, contradictions[]
POST	/api/v1/bidders/parse-gst	OCR GST REG-06 PDF → extract fields + flags	gstln, registration_type, composition_fail
POST	/api/v1/bidders/parse-udyam	OCR Udyam cert PDF → extract fields + flags	udyam_number, enterprise_classification, is_micro
POST	/api/v1/tenders/translate-regional	PyMuPDF / Bhashini pipeline for regional PDF	translation_source, translated_text_preview, rule_count
GET	/api/v1/tenders/{id}/collusion-signals	Jaccard cross-bidder cartelization scan	signals[], similarity_scores

POST	/api/v1/audit/log	Append SHA-256 audit entry	entry_id, sha256_hash, timestamp
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5. Data Requirements

5.1 Demo Bidder Profiles (mock_dataset.json)

Bidder ID	Scenario	Embedded Violation	Expected Outcome
bidder-acme-001	✔ Compliant baseline	None	Score: 92, Risk: LOW
bidder-delta-004	✘ MCA21 turnover fraud	Claimed ₹8.5 Cr, MCA21 says ₹14.5 Cr	Score: 45, Risk: HIGH + CONFLICT
bidder-theta-007	✘ Time-travel fraud	GST filed before company registration	Score: 20, Risk: CRITICAL
bidder-echo-005	✘ Multi-rule breach	MII 30% + Sub 45% + GST 6mo gap	Score: 15, Risk: CRITICAL
bidder-foxtrot-006	⚠ Partial fail	MII 35% + GST 3mo missed	Score: 58, Risk: MEDIUM

5.2 Demo Government Document PDFs

File	Type	Deliberate Violation
gst_tender_demo.pdf	GST REG-06	Registration Type: Composition (B2G ineligible)
udyam_tender_demo.pdf	Udyam Certificate	Classification: SMALL (claiming MICRO benefits)
bidder_foxtrot.pdf	Bidder Tech Bid	Local Content 35% + GST 3-month gap
telugu_tender_demo.pdf	Telugu Tender (Playwright)	All 6 rules in Telugu — for Bhashini pipeline testing

5.3 Zustand Store Schema

```
// dashboardStore.js – all global state slices
{
  selectedDocument: 'tender' | 'gst' | 'udyam',
  tenderRules: RuleObject[], // dynamically extracted
  verifiedDocResult: VerificationResult | null,
```



```
gstParseResult:      GstParseResult | null,
udyamParseResult:    UdyamParseResult | null,
auditLog:            AuditEntry[],          // SHA-256 entries
viewerRef:           { goToPage(n: number): void } | null,

// Actions
setTenderRules, setVerifiedDocResult, clearVerifiedDocResult,
setGstParseResult, clearGstParseResult,
setUdyamParseResult, clearUdyamParseResult,
addAuditEntry, setViewerRef
}
```

6. UI / UX Requirements

Feature	Requirement	Implementation
Resizable Split Panel	Drag divider left/right to resize PDF vs Dashboard panels	mousemove on window, getBoundingClientRect, empty [] deps (no stale closure)
Iframe Drag Shield	Mouse events must not be stolen by PDF iframe during drag	position:absolute; inset:0; z-index:50 overlay during isDragging
Minimize PDF Panel	Officer can hide PDF panel to expand dashboard view	Width → 0 with CSS transition; only when !isDragging
Lenis Smooth Scroll	Dashboard scrolls with physics inertia	new Lenis({ wrapper: rightPanelRef }) + rAF loop
Context-aware Upload	Button label + color matches current document tab	UPLOAD_CONFIG map: tab → endpoint + field + color classes
ReactFlow Graph Layout	Columns: Bidder x=0, Claims x=250, Evidence x=560, Anchors x=870	EvidenceGraph.jsx column auto-layout algorithm
Animated Conflict Edges	Red dashed animated edges on EVIDENCE_CONFLICT connections	animated:true, style: {stroke:'#ef4444'} on edge
Anti-cartel Modal	Full-screen modal with legal guardrail disclaimer	App.jsx modal state + Jaccard results table
Responsive Layout	Works on 1280px+ screens	Tailwind flex/grid — not mobile-optimized (procurement officer use case)

7. Non-Functional Requirements

Category	Requirement	Target
Performance	OCR + graph verification latency	< 500ms per bidder

Performance	Dashboard initial load time	< 2s (Vite bundled)
Reliability	Backend uptime during demo	Uvicorn restart < 5s
Security	Audit entries	SHA-256 hash — tamper-evident
Accuracy	GSTIN regex validation	15-char pattern [0-9]{2}[A-Z]{5}[0-9]{4}[A-Z][1-9A-Z]Z[0-9A-Z]
Accuracy	Udyam number regex	UDYAM-[A-Z]{2}-[0-9]{2}-[0-9]{7}
Scalability	Mock dataset bidders	5 profiles — architecture supports N via JSON extension
Maintainability	Config bridge	config.active_tender_limits — single source of truth for all thresholds
Compliance	Officer autonomy	System is decision-support only — final qualification decision with officer (GFR 2017)
Transparency	AI explainability	All decisions backed by evidence nodes — no black-box scoring

8. PS 26100 Compliance Matrix

PS 26100 Expected Capability	Our Implementation	Status
1. Integrate with Government portals/databases	GST OCR + Udyam OCR + MCA21 mock dataset	PARTIAL
2. Verify Udyam/MSME status	parse-udyam endpoint — full OCR parse + is_micro flag	FULL
3. Verify GST registration and return filing	parse-gst endpoint — GSTIN, type, composition flag	FULL
4. Verify PAN and Income Tax compliance	PAN hard filter in graph engine (mock)	PARTIAL
5. Check Make in India local content	MII % extracted from tender + graph contradiction	FULL
6. Verify EPFO/ESIC compliance	Not implemented	GAP
7. Startup India / NSIC / OEM authorization	Not implemented	GAP
8. DigiLocker / document verification	Mock only — architecture supports OAuth plug-in	PARTIAL
9. Blacklisting and debarment status	Debarment hard filter in graph engine	FULL
10. Tender-specific compliance checks	Dynamic rule extraction → config bridge → graph engine	FULL
11. AI to identify missing/inconsistent information	Evidence graph CONFLICT edges + contradiction panel	FULL
12. Compliance Score and Risk Level	0–100 score + LOW/MEDIUM/HIGH/CRITICAL badge	FULL
13. AI-generated recommendation to officer	Contradiction cards + Show-Cause Notice generation	FULL

14. Auditable record of verification	SHA-256 tamper-evident audit trail + log UI	FULL
BONUS: Regional language tenders	Bhashini MeitY AI pipeline (Telugu/Malayalam/Hindi...)	IMPLEMENTED
BONUS: Anti-cartel detection	Jaccard cross-bidder collusion scan	IMPLEMENTED

Coverage Summary: 9 of 14 PS requirements fully implemented (64%), 3 partial (21%), 2 gaps (EPFO/ESIC + Startup India — require government API credentials outside SIH scope). Plus 2 bonus features not required by PS: Bhashini regional language and anti-cartel detection.

9. Limitations & Future Scope

9.1 Current Limitations

Limitation	Reason	Production Fix
GSTN API not called live	Requires MeitY GSTN API key (restricted)	Plug in GSTN sandbox credentials
MCA21 data is mocked	MCA21 API requires corporate registration	MCA21 REST API integration
EPFO/ESIC not implemented	Outside SIH demo scope	EPFO SHRAM portal API
Bhashini in simulation mode	BHASHINI_TOKEN env variable not set	Set BHASHINI_TOKEN from ULCA dashboard
Single officer role	No auth layer for hackathon	Add JWT auth + role-based access
Single-bidder verification	Dropdown-based, not batch	Batch API + queue with Celery

9.2 Future Scope

- Live integration: GSTN → PAN → MCA21 → EPFO → DigiLocker → Startup India → NSIC
- Multi-officer workflow with role-based approval gates
- Real-time Bhashini translation with ULCA production API token
- Batch tender evaluation — process 50+ bidders simultaneously
- LLM-based natural language recommendation generation (GPT-4/Gemini)

- Mobile-responsive officer app
- GeM portal direct integration for bidder self-submission
- DPDP-compliant PII redaction before audit storage